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A robust framework for noisy speech recognition using Frequency-Guided-Swin Transformer

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ABSTRACT

Conventional automatic speech recognition (ASR) systems often struggle to generalize across diverse and noisy environments, where background interference significantly degrades recognition accuracy. This work presents a novel approach to noisy speech recognition by combining convolutional neural networks (CNN) and Swin Transformer with frequency-guided multi-head self-Attention (FG-MSA) architectures. The proposed method addresses the challenge of recognizing speech in noisy environments, focusing on character-level transcription from noisy audio. The CNN efficiently extracts localized features, while the Swin Transformer, with its hierarchical structure and shifted window mechanism, captures both local and long-range dependencies. The FG-MSA mechanism is introduced to guide the attention mechanism toward frequency components that are most relevant for speech recognition, improving robustness in noisy conditions. The proposed method improves performance and efficiency for ASR, especially in noisy environments. Evaluated on the Aurora-2 dataset, and the noisy speech commands (NSC) dataset. The proposed CNN-FG-Swin Transformer achieved an average accuracy of 87.19% on the isolated Aurora-2 dataset, outperforming the baseline Swin Transformer by 2.49%. For all datasets, the proposed model achieved an average accuracy of 87.01%, outperforming all the compared state-of-the-art methods. On the NSC dataset at -9 dB, it achieved a word error rate (WER) of 36.20%, outperforming the end-to-end capsule network models by 8%, both DNN 38.63% and LSTM 69.09% baselines, confirming its robustness in real-world conditions.

1. Introduction

Automatic speech recognition (ASR) has seen remarkable advancements in recent years, leading to notable improvements in both accuracy and adaptability (Djeffal et al., 2023b). These developments have been instrumental in the widespread adoption of voice-enabled technologies, facilitating more intuitive interactions between humans and machines. Applications like virtual assistants and voice-controlled devices are now commonly used, highlighting the transformative impact of ASR in enhancing user experiences across various sectors (Djeffal et al., 2024a; Hamza et al., 2023). ASR systems often face challenges in maintaining reliability and accuracy in noisy environments (Hinton et al., 2012). Background noise can alter the acoustic properties of speech signals, leading to higher error rates and diminished system performance. Additionally, background noise can mask important speech cues, making it hard for the system to differentiate between similar-sounding words. It may also introduce false activations or

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misinterpretations due to overlapping speech or sudden transient sounds (e.g., car horns, alarms). Furthermore, noise can reduce the effectiveness of traditional feature extraction techniques, which rely on clean input for accurate recognition, and can compromise speaker identification and language modeling efforts. Numerous strategies have been explored by researchers to effectively address these challenges, such as advanced signal processing methods, and robust feature extraction techniques. With the emergence of deep learning (DL), models like convolutional neural networks (CNNs) and recurrent neural networks (RNNs), including Long short-term memory (LSTM) networks, have significantly improved ASR performance by automatically learning complex features and temporal patterns from data. However, even these DL-based methods face limitations in noisy environments. They can struggle to maintain accuracy when noise alters speech dynamics or when context across longer time frames is needed. These challenges highlight the need for more advanced architectures. Transformer-based architectures, known for their exceptional performance in natural language processing and computer vision tasks (Kheddar et al., 2025a; Vaswani et al., 2017; Kheddar, 2025), have also emerged as a promising solution. Their multi-head self-attention (MSA) mechanism effectively captures long-range dependencies and contextual relationships, making them ideal for tasks involving ASR.

This paper introduces a novel approach utilizing Transformer-based MSA to enhance the effectiveness of ASR systems under noisy conditions. By combining CNN and Swin Transformer architectures for noisy ASR, our work introduces an innovative method for tackling the challenges of recognizing speech in noisy environments. While CNNs are effective at extracting localized features (Djeffal et al., 2023a), the Swin Transformer excels at capturing long-range dependencies and hierarchical representations, making their combination uniquely suited for this task. Unlike existing studies that primarily focus on Swin Transformers for tasks like keyword spotting (Sun et al., 2024), speech enhancement (Jiang et al., 2023), or emotion recognition (Liao and Shen, 2023; Liu et al., 2025). By incorporating shifted window mechanisms and frequency-guided MSA (FG-MSA) in the Swin Transformer, our model is expected to enhance the handling of noisy inputs for ASR task, improving contextual feature extraction for sequence prediction.

The proposed FG-MSA enables the model to dynamically focus on critical frequency bands, effectively capturing speech-relevant features while mitigating the impact of noise. This targeted approach further boosts the model's robustness in challenging acoustic environments. This focus on ASR at the character-level transcription in noisy conditions positions this work as a distinct and original contribution to the field.

The contribution described in this work is summarized as follows :

- Introduces a novel hybrid architecture that combines CNN with a Frequency-Guided Swin Transformer (FG-Swin) to enhance ASR in noisy environments. The CNN extracts low-level acoustic features, while the Swin Transformer captures long-range dependencies, ensuring robust speech representation.
- A novel FG-MSA mechanism is proposed within the Swin Transformer block to improve the model's capacity to concentrate on pertinent frequency components, thereby improving noise robustness and speech intelligibility. This method dynamically refines attention allocation to suppress noise while preserving critical speech features.
- The proposed approach is benchmarked against leading DL models for noisy speech recognition. Using the Aurora-2 dataset in isolated words and in all dataset, as well as the noisy speech commands (NSC) dataset, performance is evaluated using key metrics such as word recognition accuracy and word error rate (WER), highlighting the advantages of the CNN-FG Swin Transformer model.
- The efficiency of the suggested approach has been validated, which contain diverse noise conditions. This evaluation ensures that the model generalizes well to real-world noisy speech scenarios.

The rest of this paper is structured as follows: Section 2 presents an overview of relevant works. Section 3 covers the background of baseline swin Trasnformer. Section 4 present the proposed CNN-FG-swin Transformer approach. The results of the experiments involving these improvements are presented in Section 5. Finally, Section 6 concludes the work and identifies possible directions for future research.

2. Related works

Noise-robust ASR has been studied for more than four decades. Li et al. (2014) provide a comprehensive review of classical techniques developed before the deep learning era. The dominant framework was the Gaussian mixture model – hidden markov model (GMM-HMM), where GMMs captured acoustic feature distributions while HMMs modeled temporal dynamics; robustness was often improved through feature compensation, model adaptation, or specialized training. Among model-based techniques, vector taylor series (VTS) compensation was used to approximate noisy speech features through nonlinear expansions, while parallel model combination (PMC) combined clean and noise models in the model domain to better simulate noisy conditions. Model adaptation strategies, such as maximum likelihood linear regression (MLLR) and maximum a posteriori (MAP), were also widely applied to adjust GMM-HMMs to mismatched environments. Additionally, missing feature theory (MFT) addressed robustness by masking unreliable regions of the spectrogram and decoding only from reliable features. Collectively, these approaches represented the foundation of classical noise-robust ASR research prior to the deep learning era. Robust ASR in noisy environments has attracted significant research interest in recent years (Hu et al., 2022). A variety of approaches have been proposed to address the challenges posed by background noise and signal distortion, ranging from deep learning-based models to the use of specialized datasets for training and evaluation. Advances in neural network architectures, such as CNNs (Rashno et al., 2019), and Transformers-based models with MSA mechanisms (Long, 2025), including the Swin Transformer (Liu et al., 2025), combined with the availability of standardized noisy speech corpora, have enabled researchers to systematically assess model performance under diverse acoustic

conditions. In addition to these architecture-oriented approaches, other complementary strategies have also been widely investigated, such as incorporating auxiliary features (e.g., articulatory, prosodic, or bottleneck representations), applying speech enhancement as a preprocessing step, and adopting noise-aware training techniques to further improve robustness. Noise-aware training (NAT) improves robustness in speech enhancement and ASR by incorporating noise information during training. Techniques include multi-condition training with diverse noise types, noise-conditioned training using noise embeddings, and data augmentation with artificial noise or reverberation. Adaptive noise suppression, domain adversarial training, and joint enhancement-recognition approaches further enhance performance. These methods help systems generalize better to challenging and unseen noisy environments. In recent years, speech enhancement has progressed from early deep learning approaches to advanced real-time generative models. Initial studies, such as Xu et al. (2014) and Wang and Chen (2018), applied convolutional and recurrent networks to map noisy speech to clean signals, improving intelligibility and robustness. Tan and Wang (2019) further advanced the field with complex spectral mapping techniques. More recent works, including DeepFilterNet (Schröter et al., 2023) and compact DNN frameworks, emphasize lightweight, unsupervised, and end-to-end strategies for robust noise reduction. Diffusion-based and normalizing flow models have also emerged, offering powerful noise suppression without paired training data. These innovations make speech enhancement more adaptable and effective in complex, real-world noisy conditions. Table 1 Summary of related works using Swin Transformer and CNN for speech recognition and noise-robust speech recognition.

2.1. Swin transformer-based clean and noisy speech processing

The Swin Transformer has gained significant attention in research. Recent works have also explored its applications and advancements in speech processing. Sun et al. (2024) address the challenge of high computational demands in Transformer-based keyword spotting (KWS) models, which hinders their use in resource-limited environments. To address this, they propose an innovative method that utilizes a Swin-Transformer architecture to balance efficiency and accuracy. The process begins with the extraction of mel-frequency cepstral coefficients (MFCCs) from speech signals, followed by the use of a temporal convolutional network (TCN) to extract dynamic features. Additionally, a frame-level shifted window attention mechanism is introduced to enable interaction between adjacent windows, enhancing the capacity of the model to capture contextual and time–frequency features. Testing on the Speech Commands V1 dataset reveals that this method outperforming existing KWS techniques while maintaining a lightweight design. However, the model is evaluated only under clean conditions and does not assess performance in noisy or real-world environments. Jiang et al. (2023), introduce a hierarchical frame-level Swin Transformer network (FLSTN) to address the computational challenges of using Transformers for speech enhancement, particularly in noisy environments. By limiting self-attention to localized windows of consecutive frames, they effectively reduce the computational complexity from quadratic to linear relative to the size of the input spectrogram. To ensure information exchange between adjacent windows, they implement a shifted window mechanism, creating an effect similar to global attention. The hierarchical design of FLSTN allows the model to extract features at multiple scales, enhancing its ability to represent speech. Additionally, they develop specialized layers band merging to reduce spatial resolution and band expanding to further optimize computational efficiency without sacrificing performance. The proposed approach is tested on both 16 kHz and 48 kHz speech datasets, demonstrating robust performance across different bandwidths. Nonetheless, the model's complexity may still pose challenges for real-time deployment on low-power devices. The evaluation does not explicitly address robustness under real-world noisy conditions, which limits insight into its generalization to diverse acoustic environments. Liao and Shen (2023), propose a speech emotion recognition (SER) system based on the Swin Transformer, leveraging its capabilities to process spectrogram representations of audio signals. They explore the viability of using the Swin Transformer for SER tasks and compare the performance of different spectrogram types within the same model framework to evaluate their effectiveness. However, the study primarily focuses on model feasibility rather than optimizing for generalization across diverse emotional datasets, which may affect its adaptability to varied speech emotion contexts. The work is limited to emotional speech classification under clean conditions, without examining the impact of background noise or acoustic interference. Liu et al. (2025), introduce ESERNet, a Swin Transformer-based architecture for SER that successfully extracts long-range dependencies and crucial emotional cues from speech signals. Leveraging a dual-pathway structure applied to mel-spectrograms, the model addresses key challenges such as emotional ambiguity and the existence of background noise. However, ESERNet is not explicitly tailored for noisy ASR, its robustness to noise makes it well-suited for practical SER scenarios involving real-world acoustic variability.

2.2. Transformers-based noisy speech recognition

Self-supervised learning (SSL) has become a potent framework for speech processing (Chang et al., 2022). Models such as Wav2Vec 2.0 (Baevski et al., 2020; Zhu et al., 2022) and HuBERT (Hsu et al., 2021) learn rich speech representations from raw audio through contrastive or masked prediction objectives. These models are pretrained on large corpora and then fine-tuned on downstream tasks such as ASR. HuBERT, in particular, improves upon Wav2Vec 2.0 by predicting masked cluster assignments rather than continuous features, yielding better performance on some benchmarks. Building on this line of work, in Ng et al. (2023) propose a reduction principle in the HuBERT training framework. Their method incorporates auxiliary loss functions that guide the self and cross correlation matrices of noise-affected embeddings toward identity matrices. This design prompts the model to learn noise-invariant speech representations. As a result, the framework demonstrates enhanced recognition performance in a range of noisy environments, including previously unseen noise conditions, while maintaining accuracy in clean speech data. Wang et al. (2022), present wav2vec-Switch, this approach introduces a novel strategy, it simultaneously processes clean and noisy versions of

Table 1
Summary of related works employing Swin Transformers and CNNs across various speech tasks, including noise-robust speech recognition.

	Ref	Year	Models	Datasets	Focus/noise types	Metrics	Performance
Swin and/or CNN	Liu et al. (2025)	2025	Swin-ESERNet	IEMOCAP, EmoDB	Speech emotion recognition (noisy)	Accuracy	76.85%, 93.85%
	Sun et al. (2024)	2024	Swin-Transformer, TCN	Speech Commands V1	Keyword spotting (clean audio)	Accuracy	98.01%
	Jiang et al. (2023)	2023	Swin-FLSTN	WSJ0, VoiceBank-DEMAND	Speech enhancement (no specific noise)	PESQ, STOI, CSIG, CBAK, COVL	2.88*, 0.944, 4.18, 3.43, 3.53
	Liao and Shen (2023)	2023	Swin Transformer	IEMOCAP	Speech emotion recognition (clean)	Accuracy	70.1% WA, 62.4% UA
	Rashno et al. (2019)	2019	NCNN	Aurora2	Noise-robust ASR	Accuracy	85.96%
	Qian et al. (2016)	2016	CNN	Aurora4	Noise-robust ASR	WER	8.81%
Noise-robust speech recognition	Radford et al. (2023)	2023	OpenAI Whisper	CHiME-6, LibriSpeech other, Tedlium 3	Overlapping speech, audiobook, udience laughter	WER	5.2%, 4.0%, 25.5%
	Chen et al. (2022)	2022	WavLM	LibriSpeech test-other	Audiobook recordings	WER	4.0%
	Wang et al. (2022)	2022	Wav2Vec-Switch	CHiME-4	Bus, cafe, pedestrian, street	WER	6.6%
	Nasersharif and Naderi (2021)	2021	CBN	Aurora2	Subway, car, exhibition	Accuracy	26.89%
	Nasret et al. (2021)	2021	Neural VAD	Aurora2, Aurora3	Mixed noise types	Error Rate	10.3%
	Fan et al. (2020)	2021	GRF	AISHELL-1	Traffic, claps, animals	CER reduction	10.04%
	Soe Naing et al. (2020)	2020	DWD	Aurora2	Subway, car, exhibition	Accuracy	44.55%
	Pervaiz et al. (2020)	2020	DNN, LSTM, CNN	Speech Commands	Taps, pink/white noise	Top-One error	88.2%
	Quintanilha et al. (2020)	2020	DNN	LibriSpeech, BRSD	Clean speech	CER, WER	10.49%, 24.45%
	Shimada et al. (2019)	2019	MVDR, MNMF	CHiME-3	Urban noises	WER	13.88–16.16%
	Bae and Kim (2018)	2018	Capsule network	Speech Commands	Taps, dishwashing, white/pink noise	CER reduction	51.7%
	Wang et al. (2018)	2018	DNN	Custom utterances	Reverberation, interference	WER	6.56%
	Candido Junior et al. (2023)	2023	DNN	Common Voice	Spontaneous, communication noise	WER	24.18%, 20.08%
	Duarte and Colcher (2024)	2024	DNN	Common Voice	Communication interference	CER	31.64% (SNR>0), 23.00% (SNR=30)
	This work	2025	FG-Swin Transformer	Aurora2, Speech Commands	Subway, babble, car, pink, white noise	Accuracy, WER	87.19%, 36.20%

Note: * = Perceptual Evaluation of Speech Quality (PESQ). WA = Weighted Accuracy. UA = Unweighted Accuracy.

the same audio input and swaps their quantized representations during training. This encourages the model to generate consistent outputs for both conditions, promoting noise-invariant learning. Experimental results show that wav2vec-Switch improves WER on both synthetic noisy LibriSpeech and real-world noisy datasets, including the CHiME-4 1-channel corpus, without compromising accuracy on clean speech, outperforming standard data augmentation and even some speech enhancement systems. While SSL models like Wav2Vec 2.0, HuBERT, and deHuBERT achieve impressive results on continuous speech datasets such as LibriSpeech and CHiME-4, they require extensive computational resources and large-scale pretraining, making them less practical for low-resource or real-time settings, notably under severe noise conditions. Radford et al. (2023) introduced large-scale speech recognition models trained on approximately 680,000 h of multilingual and multitask audio–text data gathered from the internet. Instead of relying on carefully annotated corpora, they adopted a weakly supervised approach in which transcripts contained noise and imperfections. Owing to the vast scale of the dataset, the models were able to acquire robust representations that transferred effectively to a wide range of ASR benchmarks. Their systems achieved performance comparable to fully supervised methods and, in some cases, even surpassed them, all without requiring dataset-specific fine-tuning. Furthermore, the models demonstrated resilience close to human-level accuracy, and both the trained models and inference code were released to encourage further advancements in robust speech processing. Chen et al. (2022) proposed WavLM, a self-supervised model aimed at addressing a broad range of speech processing tasks, not just automatic speech recognition. Their approach integrates masked speech prediction with a denoising objective during pre-training, allowing the model to capture linguistic content while also becoming more robust for tasks beyond ASR. To improve sequential modeling, they modified the Transformer architecture by incorporating a gated relative position bias. The training corpus was also expanded from 60,000 to 94,000 h of audio to further enhance generalization. Experimental results showed that WavLM, particularly its large version, achieved state-of-the-art performance on the SUPERB benchmark and delivered notable gains across multiple speech processing tasks.

2.3. Noisy speech recognition using the Aurora-2 dataset

The Aurora-2 dataset is widely used by researchers in noisy speech recognition due to its extensive variety of challenging noise conditions. It serves as a benchmark for evaluating speech recognition systems performance under various signal-to-noise ratios (SNRs). Soe Naing et al. (2020), address the challenge of improving ASR systems in noisy environments by proposing a framework that integrates discrete wavelet denoising (DWD) with feature extraction. The authors explore various wavelet families and thresholding techniques to lessen noise interference and enhance the robustness of MFCCs. The framework's effectiveness is evaluated using GMM-HMM and deep neural network-hidden markov model (DNN-HMM) as recognizers. Results on the Aurora-2 dataset show that combining wavelet denoising, specifically using the Coiflet wavelet and Rigrsure threshold, with cross-entropy-trained DNN-HMM achieves the best recognition accuracy, particularly at higher SNRs. However, the approach shows diminished effectiveness at lower SNR levels, where noise severely distorts the speech signal. Nasersharif and Naderi (2021), explore the use of bottleneck features extracted by convolutive bottleneck networks (CBNs), a variant of CNNs, for improving noisy speech recognition. The authors highlight that CBNs include a bottleneck layer within their fully connected layers, enabling the extraction of discriminative features with rich contextual information. From an information theory perspective, the study demonstrates that the information exchanged between the bottleneck features and the output labels is higher compared to that of the noisy input features and labels. This indicates that bottleneck features provide a denoised and compressed representation that is more effective for distinguishing phone classes. The proposed method processes noisy logarithmic Mel filter bank energies (LMFBs) from neighboring frames as inputs to the CBN, with outputs corresponding to phone labels. Experimental evaluation on the Aurora-2 dataset reveals that these bottleneck features outperform traditional speech features and robust features extracted by standard CNNs, demonstrating their effectiveness in handling noisy environments. Nevertheless, the method relies on carefully selected input contexts and may require extensive tuning to generalize across different noise types or real-world scenarios, especially under low SNR conditions. Fazel and Chakrabartty (2011), proposed sparse auditory reproducing kernel (SPARK) coefficients for noise-robust speech feature extraction. This method uses kernel-based similarities with gammatone functions and max pooling to capture noise-resilient information. Experiments on the Aurora-2 dataset showed improved recognition over standard ETSI features. However, the approach lacks adaptive feature learning and global context modeling, limiting its scalability in real-world noisy conditions.

2.4. Noisy speech recognition using the speech commands dataset

The Speech Commands (SC) dataset is a widely used standard for assessing speech recognition models in command-based applications. For example, Pervaiz et al. (2020) investigate advancements in speech recognition by analyzing the impact of modern devices, emerging technologies, and machine learning techniques. Their study conducts an in-depth comparison of various machine learning (ML) and DL models to determine the most efficient approach for robust speech recognition in noisy conditions. The researchers developed and tested multiple models, including traditional methods like Gaussian mixture models combined with GMM-HMM, as well as deep learning-based approaches such as deep neural networks (DNNs), CNNs, and LSTMs. To assess the impact of noise, the models were evaluated using noisy audio data, and the effect of training on noisy inputs was analyzed. In addition to background noise, the study addresses other challenges in speech recognition such as accent variations, dialect differences, code-switching (mixed language), and non-native speakers by creating a custom dataset collected locally. This dataset was used to test the models' performance in more realistic, real-time scenarios. However, the study lacks an in-depth evaluation of model generalization across unseen noise types, which is essential for ensuring robustness in diverse acoustic environments. Bae and Kim (2018), explore speech recognition techniques, particularly models utilizing CNNs and RNNs, which have demonstrated strong

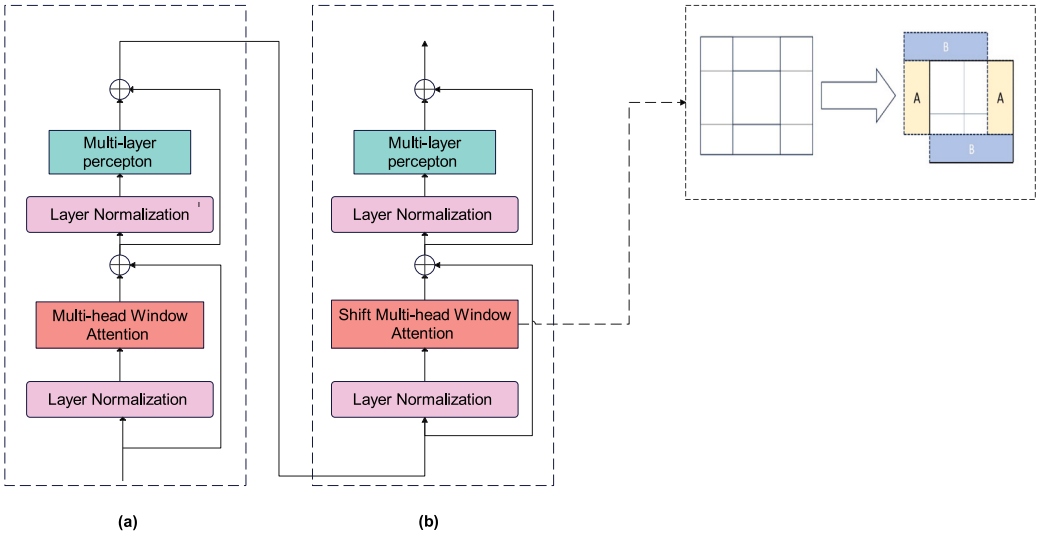


Fig. 1. The architecture of baseline Swin Transformer: (a) Swin Block (b) Shift Swin Block (Sun et al., 2024).

performance in various benchmarks. While CNNs effectively extract local features, they are mainly suited for tasks with short-term dependencies, such as keyword detection and phoneme recognition. However, a drawback of CNNs is that max-pooling can ignore the spatial relationships between low-level features. To overcome this limitation, the study introduces capsule networks, which preserve spatial and pose information within speech spectrograms across both frequency and time dimensions. Their end-to-end model, tested under both clean and noisy conditions. The findings indicate that this approach surpasses traditional CNN-based models, leading to improved speech recognition accuracy. Nonetheless, capsule networks require longer training times and more complex tuning compared to conventional CNNs, which may hinder their scalability in large-scale deployments.

3. Background

3.1. Baseline swin transformer block

The Swin Transformer block in the architecture alternates between window-based MSA (W-MSA) and shifted window-based MSA (SW-MSA) (Park et al., 2025) as illustrated in Fig. 1. The SW-MSA module introduces a shift mechanism applied to the feature maps using a custom ShiftedWindowLayer, which shifts the input tensor spatially before applying MSA. This shift enables cross-window interactions, complementing the local attention in W-MSA (Sun et al., 2024). In practice, a Swin Transformer block consists of a shifted window-based MSA module, followed by a two-layer MLP with a GELU activation in between. Layer normalization (LN) is applied before each MSA and MLP, while residual connections are added after every module to ensure stable training. Unlike the standard Transformer, which relies on global self-attention and incurs quadratic complexity with the number of tokens, the Swin Transformer limits attention to non-overlapping windows, significantly lowering computational cost. By alternating between W-MSA and SW-MSA, it maintains efficiency for high-resolution inputs while still capturing both local patterns and cross-window dependencies (Liu et al., 2021).

MSA enables the model to attend to multiple representation subspaces simultaneously, capturing information from various positions within the input. Its general formulation is expressed as follows Vaswani et al. (2017): $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O$, where each head_i is computed as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

. The projection matrices are $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

The MSA mechanism is computed as Zhang et al. (2021):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

Where Q, K, V are the query, key, and value matrices, d is the query/key dimension. Multiple attention heads operate in parallel to capture diverse representations (Djeffal et al., 2024b).

3.2. Windowed attention

The windowed self-attention mechanism offers a substantial decrease in computational complexity compared to the conventional self-attention approach. However, its design confines attention to individual windows, limiting inter-window connections and reducing the extent of information exchange. To improve global feature extraction, the shift-Swin module incorporates a windowed self-attention mechanism with a shifted window operation. In this mechanism, the window shifts by half its size during the sliding process, enhancing information exchange between adjacent windows. The shifted window operation creates empty regions at the edges of the feature map. To address this, feature information from neighboring windows is copied into these empty regions, and the filled areas are then masked to maintain consistency. This effectively resolves the issue of feature vacancies (Sun et al., 2024).

Additionally, a residual connection is integrated into each Swin module. This addition helps prevent gradient vanishing and further enhances the module's capability to extract local features. The shifted window operation is illustrated in Fig. 1.

3.3. CTC

The connectionist temporal classification (CTC) loss function, which enables the model to learn the alignment between the input audio sequence and the corresponding target transcription without needing a precomputed alignment (Graves et al., 2006). The probability of a transcription Y given an input sequence X is computed by summing over all possible alignment paths that, after removing blank tokens and collapsing repeated symbols, yield the target transcription Y:

$$p(Y \mid X) = \sum_{\pi \in B^{-1}(Y)} \prod_{t=1}^{T} p(\pi_t \mid X) \tag{2}$$

Where, π represents a potential alignment path, $B^{-1}(Y)$ denotes the set of all alignment paths that collapse to Y, and $p(\pi_t \mid X)$ represents the probability of symbol π_t at time step t. The CTC loss function is defined as the negative logarithm of this probability:

$$L_{\text{CTC}} = -\log p(Y \mid X) \tag{3}$$

This method handles temporal variability in speech (Djeffal et al., 2023b), allowing flexible alignment between the acoustic input sequence and the target transcription.

4. Proposed approach

The core innovation of the proposed architecture lies in the integration of FG-Swin Transformer blocks, which extend the traditional Swin Transformer by incorporating FG-MSA. This architecture is designed for ASR in noisy environments. The architecture processes input spectrograms through CNNs for initial feature extraction and FG-Swin Transformer blocks with W-MSA, SW-MSA, FG-MSA and multi-layer perceptrons (MLPs) to enhance contextual understanding, as shown in Fig. 2.

4.1. Mel-spectrogram extraction

The proposed CNN model extracts Mel-spectrogram features from audio signals using parameters tailored to achieve robust and efficient feature representation (Meghanani et al., 2021). Each audio file is processed with a hamming window of 25 ms (200 samples at a sampling rate of 8 kHz) and a hop length of 10 ms (80 samples), ensuring an appropriate overlap between frames for detailed temporal analysis. The fast Fourier transform computation is performed with 2048 points, providing high-frequency resolution necessary for capturing intricate patterns in the signal. The Mel-spectrogram incorporates 40 Mel filter banks, spanning frequencies from 50 Hz to 4000 Hz, a range well-suited for human speech.

To avoid numerical instability during logarithmic scaling, a small epsilon value is added to the Mel spectrogram. This scaling transforms the spectrogram into decibels, emphasizing perceptually significant variations while compressing the dynamic range. These preprocessing steps ensure that the extracted features capture both time and frequency domain information critical for speech recognition tasks, particularly in noisy environments. This feature representation is fed into the CNN model, leveraging its ability to learn hierarchical patterns for robust performance.

4.2. Frequency guided-swin transformer

The proposed FG-Swin Transformer model applies a series of three Swin Transformer blocks, as illustrated in Fig. 3, incorporating W-MSA, SW-MSA, and FG-MSA mechanisms. In the W-MSA block, standard window-based self-attention is used. In the SW-MSA block, the shifted window multi-head self-attention mechanism is introduced to effectively capture long-range dependencies in the feature maps. This shifting strategy enhances robustness to high levels of noise by reducing the risk of over-focusing on isolated regions. Each Swin Transformer block is followed by an MLP with ReLU activation, enabling deeper and more complex feature transformations. Additionally, residual connections within each block help stabilize the training process.

To further improve performance in noisy environments, the final Swin Transformer block includes a frequency-guided MSA module. FG-MSA computes frequency-aware weights by averaging features over time to emphasize key spectral components. These weights guide attention toward important frequencies, improving speech recognition in noisy conditions.

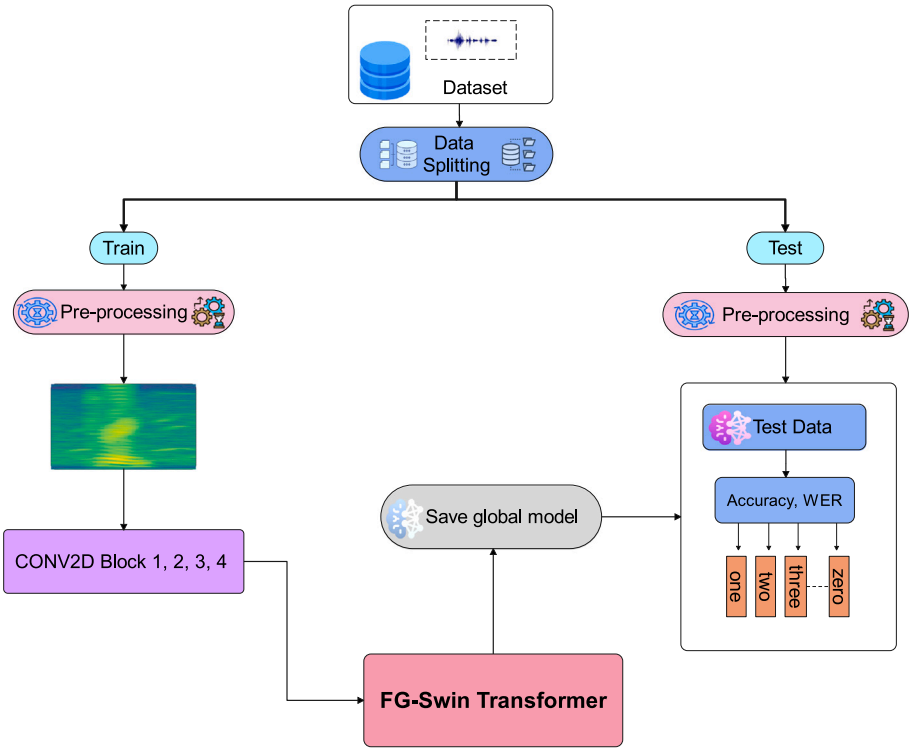


Fig. 2. The proposed architecture CNN-FG-Swin Transformer for efficient ASR applied to noisy dataset.

4.2.1. W-MSA

W-MSA is a key component of the Swin Transformer architecture, designed to tackle the computational limitations of traditional global self-attention mechanisms when dealing with high-resolution inputs. In standard self-attention (Wang et al., 2024), the computational cost grows quadratically with the input size, which becomes inefficient and impractical for large-scale feature maps. W-MSA overcomes this by dividing the input feature map into a set of fixed-size, non-overlapping windows and performing self-attention separately within each window (Liu et al., 2021). This local attention mechanism ensures that computational complexity is reduced to a linear scale relative to input size, making it significantly more efficient. Despite the localization, W-MSA retains the benefits of multi-head attention by enabling the model to jointly attend to information from various representation subspaces within each window. In the proposed model, W-MSA is implemented through a Swin Transformer block where the $shift_size$ parameter is set to zero, indicating that no window shift is applied. This configuration allows the model to concentrate on learning detailed, fine-grained features within each localized region of the spectrogram, which is particularly important in audio and speech-related tasks where local frequency-temporal patterns carry essential phonetic information. Eq. (4) represents the standard W-MSA mechanism used in a Swin Transformer block (Liu et al., 2021).

$$\begin{cases} \hat{z}_l = \text{W-MSA}(\text{LN}(z_{l-1})) + z_{l-1}, \\ z_l = \text{MLP}(\text{LN}(\hat{z}_l)) + \hat{z}_l. \end{cases} \tag{4}$$

The input feature z_{l-1} is first normalized using normalization layer (NL), then processed by the W-MSA module, which performs self-attention within non-overlapping local windows. A residual connection is added by summing the result of the attention operation with the original input, producing an intermediate output $\hat{z}_l$. This intermediate result is further normalized and passed through a MLP, followed by another residual connection. The final output of this first sub-block is denoted as z_l. This structure enables the model to capture rich contextual relationships within local windows efficiently.

Since the self-attention mechanism is inherently permutation-invariant and does not encode the order of tokens, positional information must be added to the input embeddings. In the standard Transformer, this is achieved through positional encodings defined as follows Vaswani et al. (2017):

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \quad PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \tag{5}$$

where pos is the position index, i is the dimension, and d_{model} is the embedding dimension. These encodings are added to the token embeddings to provide the model with sequence order information.

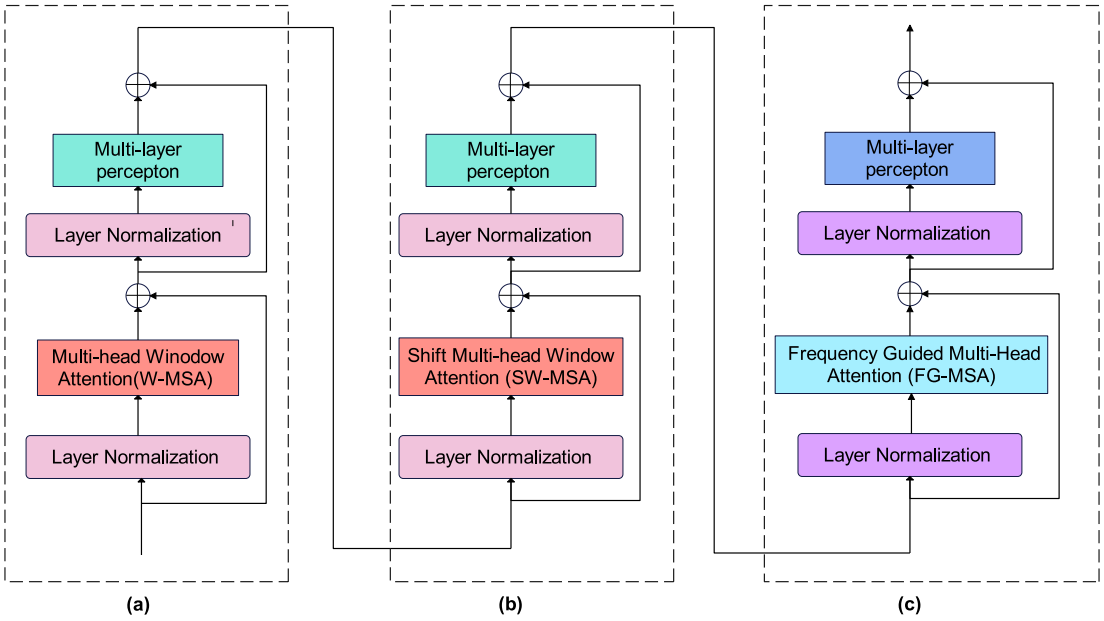


Fig. 3. The proposed architecture of FG-Swin Transformer: (a) Swin block (b) Shift Swin block (c) Frequency-Guided Swin block.

Unlike the original Transformer, the Swin Transformer employs a *relative positional bias*, which assigns learnable parameters to relative positions within a window, thereby improving modeling of local spatial relationships in high-resolution inputs (Liu et al., 2021).

4.2.2. SW-MSA

SW-MSA is an enhancement of W-MSA that addresses one of its primary limitations, which is the inability to model interactions across the boundaries of non-overlapping windows (Liu et al., 2021). While W-MSA is efficient, its local nature restricts the scope of information exchange, which can hinder the model's ability to understand more global or long-range dependencies. SW-MSA introduces a clever yet simple strategy to overcome this constraint by applying a fixed-size shift to the window partitions before computing self-attention (Wang et al., 2024). For example, in the proposed model, the second Swin Transformer block is configured with a $shift_size = 2$, meaning that the windows are shifted by two pixels both horizontally and vertically. This shift results in overlapping regions between the newly formed windows and the original ones, effectively enabling cross-window communication. By alternating layers of W-MSA and SW-MSA, the architecture ensures that information can flow seamlessly across the entire spatial domain of the input (Mahum et al., 2025). This layered attention design captures both local detail and global structure, enhancing the model's representational power. In the context of noisy speech recognition, SW-MSA plays an important part in allowing the network to integrate contextual cues from adjacent time-frequency regions, thereby improving robustness to distortions and noise present in the audio signal. Eq. (6) describes the SW-MSA sub-block (Liu et al., 2021).

$$\begin{cases} \hat{z}_{l+1} = \text{SW-MSA}(\text{LN}(z_l)) + z_l, \\ z_{l+1} = \text{MLP}(\text{LN}(\hat{z}_{l+1})) + \hat{z}_{l+1}. \end{cases} \tag{6}$$

This builds upon the output z_l from the previous step, where the feature z_l undergoes NL and is passed through the SW-MSA module, in which the windows are shifted relative to the previous layer. This shifting allows the model to improve global context modeling and capture cross-window dependencies. The attention output is added back to the input z_l via a residual connection, resulting in $\hat{z}_{l+1}$. This is then normalized and processed by an MLP, and another residual connection is utilized to obtain the final output z_{l+1} of this block. Together, these two sub-blocks form one full Swin Transformer layer, enabling both local and global feature learning.

In the Swin Transformer architecture, explicit absolute positional encodings, such as sinusoidal or learnable embeddings used in the original Transformer (Vaswani et al., 2017), are replaced by a relative position bias mechanism. This approach is particularly integrated into the SW-MSA module, where each query-key pair within a window receives an additional learnable bias based on their relative spatial displacement. Formally, the attention calculation is expressed as Liu et al. (2021):

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \tag{7}$$

where $Q, K, V \in \mathbb{R}^{M^2 \times d}$ are the query, key, and value matrices of a window of size $M \times M$, d is the dimension of queries and keys, and $B \in \mathbb{R}^{M^2 \times M^2}$ is the relative position bias matrix. Instead of encoding absolute positions, the bias values are parameterized

as a learnable table $\hat{B} \in \mathbb{R}^{(2M-1)\times(2M-1)}$, which is indexed according to the relative coordinates of token pairs. This design allows the model to encode spatial relationships efficiently while remaining flexible to variable input sizes. The use of relative position bias in SW-MSA not only reduces the need for fixed positional embeddings but also enhances the model's ability to capture local frequency-temporal dependencies in spectrograms for speech recognition tasks (Liu et al., 2021).

4.2.3. FG-MSA

FG-MSA is an advanced variant of the traditional MSA mechanism, specifically designed to leverage frequency-domain features, which is crucial for tasks like speech recognition and audio processing. In standard MSA, attention is computed by evaluating the relationships between input tokens across the sequence, without explicit consideration of the underlying frequency characteristics of the data (Vaswani et al., 2017). To further enhance performance, FG-MSA is integrated into the baseline Swin Transformer architecture, enabling the model to effectively exploit frequency-guided attention in a hierarchical and window-based structure.

FG-MSA introduces frequency embeddings or frequency biases into the attention scores, enabling the model to prioritize certain frequency components over others. This improves the capacity of the model to handle noisy environments by focusing on the most informative parts of the signal and better capturing speech-specific patterns (Pons Puig and Serra, 2018). As a result, FG-MSA significantly enhances performance in tasks like noisy speech recognition, where distinguishing between the signal and noise across various frequencies is essential (Zhang et al., 2022). Eq. (8) defines the structure of a transformer block that integrates FG-MSA.

$$\begin{cases} \hat{z}_{l+1} = \text{FG-MSA}(\text{LN}(z_l)) + z_l, \\ z_{l+1} = \text{MLP}(\text{LN}(\hat{z}_{l+1})) + \hat{z}_{l+1}. \end{cases} \tag{8}$$

The input z_l is first normalized using NL, then passed through the FG-MSA module. The output of the attention operation is added back to the original input z_l using a residual connection, yielding an intermediate result $\hat{z}_{l+1}$. This intermediate output is normalized and passed through a MLP, followed by another residual connection to produce the final output z_l. This structure is designed to improve the model's capacity to extract frequency-aware representations by modifying the attention computation.

To enhance the attention mechanism with frequency-awareness, we introduce a FG-MSA module. In this design, the additive bias term $\mathbf{F}$ in Eq. (10) is not a static parameter but is dynamically computed from the input itself. Specifically, given the input spectrogram tensor $\mathbf{X} \in \mathbb{R}^{B\times F\times T\times 1}$, where B is the batch size, F the number of frequency bands, and T the number of time steps, we compute a frequency-guided weighting map as:

$$\mathbf{F} = \text{mean}(\mathbf{X}, \text{axis} = 2, \text{keepdims} = \text{True}) \tag{9}$$

This operation aggregates energy across time for each frequency band, resulting in a tensor $\mathbf{F} \in \mathbb{R}^{B\times F\times 1\times 1}$, which highlights frequency bands with higher overall activation. This frequency-guided map is then broadcast and used to modulate the original input via element-wise multiplication $\mathbf{X}' = \mathbf{X} \odot \mathbf{F}$. The modulated tensor $\mathbf{X}'$ is then passed to a Multi-Head Attention module, and the attention mechanism can be formulated as:

$$\text{FG_MSA}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}} + F\right)V \tag{10}$$

This process biases the attention scores toward frequency components that are more prominent, effectively incorporating spectral context into the attention mechanism without requiring explicit Fourier or DCT transforms. This technique enables the model to better focus on speech-relevant spectral patterns, especially in noisy environments, by guiding attention based on intrinsic frequency saliency.

4.3. Architecture overview

The proposed CNN-FG-Swin Transformer is applied to both isolated words and the all dataset, combining continuous speech with isolated words. The model begins with a sequence of convolutional layers, each designed to capture progressively complex features from the input spectrograms. These layers include filters of increasing sizes 64, 128, 256, and 512 filters helping the model learn nuanced speech patterns essential for ASR. Each Conv2D layer is followed by normalization and max pooling, refining and condensing the feature maps while making the model more robust to noise. Following the CNN layers, the model incorporates three Swin Transformer blocks W-MSA, SW-MSA, and FG-MSA. Each block also includes an MLP layer with ReLU activation and uses residual connections to support efficient feature learning and stable training. In the final block, a FG-MSA mechanism is applied.

After passing through the Swin Transformer blocks, in isolated words the model uses the following global average pooling:

$$z_c = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} X_{h,w,c} \tag{11}$$

This latter is used to condense the spatial dimensions into a single vector as detailed in Eq. (11), which is then reshaped to match the sequence length needed for ASR $Z \in \mathbb{R}^{\text{seq_len}\times d}$. This is followed by time-distributed dense layers $P(y_t \mid X) = \text{Softmax}(W h_t + b)$ that learn temporal dependencies, essential for accurately capturing the dynamics of spoken words. For isolated words, the model was compiled using the Adam optimizer and the categorical cross-entropy loss function to optimize training performance. In the all dataset (which encompasses both isolated words and continuous speech), the model employs a bidirectional long short-term memory (BiLSTM) layer to effectively capture sequential dependencies in continuous speech recognition. The CTC is essential for recognizing continuous speech, training the model to learn variable-length sequences without requiring alignment between input frames and target labels. This modification enhances the model's ability to recognize continuous words sequences while preserving the original feature extraction capabilities of the CNN-FG Swin Transformer. This architecture synergizes CNN feature extraction, Shifted MSA, and MLPs to achieve robust ASR performance, particularly effective in noisy environments. The architecture is illustrated in Fig. 2.

Table 2
Aurora-2 dataset details.

Category	Content
Isolated words	one, two, three, four, five, six, seven, eight, nine, zero, oh
Continuous speech	zero two three one nine zero nine one three oh two eight nine oh zero four four three six five nine one three five oh six six seven five seven two one five one nine oh... etc.
Noise environments	Set A: subway, babble, car, exhibition Set B: restaurant, street, airport, station Set C: subway (MIRS), street (MIRS)

Table 3
SC dataset details.

Command	Background noise
Main commands C1 down, eight, five, four, go, left, nine, no, off, on, one, right, seven, six, stop, three, two, up, yes, zero	 doing_the_dishes dude_miaowing exercise_bike
Sub-commands C2 bed, bird, cat, dog, happy, house, marvin, sheila, tree, wow	pink_noise running_tap white_noise

5. Experiments

To evaluate the effectiveness of the proposed model in noisy speech recognition tasks, a series of experiments were conducted using standard benchmark datasets. The experiments are designed to assess performance under various noise conditions, speaker variability, and SNR levels. We compare our approach with baseline models to demonstrate improvements in robustness and accuracy across multiple test scenarios.

5.1. Datasets and metrics

5.1.1. Aurora-2 dataset

The Aurora-2 dataset was developed to evaluate speech recognition algorithms in noisy environments. It is based on the TIDigits dataset and consists of eleven connected digit utterances spoken by American English speakers, with audio downsampled to 8 kHz (Soe Naing et al., 2020). The all dataset includes isolated words and continuous speech, as presented in Table 2. It provides two training configurations: one with clean data and another with a mix of clean and noisy data. In the clean training mode, 8440 utterances are included, each processed with G.712 filtering. The multi-condition training mode divides the same utterances into 20 subsets, simulating four types of noise: suburban train, babble, car, and exhibition hall, at five SNR levels of 20 dB, 15 dB, 10 dB, 5 dB, and clean (Naing et al., 2020). The testing phase uses three test sets containing 4004 utterances in total. These utterances are evenly distributed between 52 male and 52 female speakers, with 1001 utterances per test set. Noise is introduced at SNRs ranging from 20 dB to −5 dB, in addition to clean conditions. The test sets are categorized into three groups Set A, Set B, and Set C based on different noise (Hirsch, 2000), as described in Table 2. For isolated words the dataset consists of 4824 isolated word files from 8440, with 2412 files allocated to each mode. For the test sets, 298 isolated word files were selected from the 1001 sentence-based files in each subset, resulting in a total of 2086 files per test, calculated from 298 files in each of the 7 folders named Clean and 20 dB to −5 dB.

5.1.2. Speech commands dataset

The SC dataset version 1 consists of 30 commands, totaling 64,727 utterances recorded by 1881 speakers. Each audio sample lasts one second, containing a single utterance, and is recorded in WAV format at a 16 kHz sampling rate (Pervaiz et al., 2020). Additionally, six types of noise were incorporated into the dataset, some sourced from real-world environments and others generated synthetically, including running tap, doing dishes, white noise, and pink noise. The duration of the noisy files exceeded that of the utterances. Table 3 provides details on the commands and noise types. These datasets were collected in a controlled environment, with most contributions coming from European speakers (Bae and Kim, 2018). In NSC dataset we perform multi-condition training and testing, similar to the setup used in the Aurora-2 dataset.

5.1.3. Metrics

- **Accuracy, Recall, Precision:** To assess the effectiveness of a proposed method in ASR, common metrics such as classification accuracy, recall (sensitivity), and precision (positive predictive value) are used (Kheddar et al., 2025b; Noureddine et al., 2023; Habchi et al., 2025). These metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

- **Word error rate (WER):** This metric calculates the proportion of incorrect words relative to the total number of words processed (Kheddar et al., 2023). It is expressed as:

$$\text{WER} = \frac{\text{S} + \text{D} + \text{I}}{\text{N}} = \frac{\text{S} + \text{D} + \text{I}}{\text{H} + \text{S} + \text{D}} \tag{14}$$

The calculation is based on counts of substitutions (S), deletions (D), insertions (I), hits (H), and the total number of words (N).

5.2. Model setting and hyperparameters

Table 4 presents the model parameters for isolated words and for all dataset (continuous speech with isolated words). The input spectrograms are 40 × 98 for isolated words and 40 × 300 for continuous speech with isolated words. The model employs the Adam optimizer with a learning rate of 0.0001, chosen for its ability to adaptively adjust learning rates for stable convergence. A dropout rate of 0.5 is applied after dense layers to mitigate overfitting, particularly in noisy environments. The training is conducted with a batch size of 32 and 50 epochs, balancing computational efficiency and the ability to learn complex patterns. for isolated words Global average pooling is applied after feature extraction blocks to reduce feature dimensionality while preserving critical information. TimeDistributed layers are integrated to handle sequence data, enabling the model to predict character-level sequences for each audio input. For continuous speech, the proposed model compresses the frequency dimension to extract distinctive representations while preserving the temporal dimension as a sequence. More precisely, the spectrogram tensor is rearranged and transformed so that each time frame is represented as an independent feature vector. The sequence is then processed using two successive BiLSTM with 256 and 128 units, respectively, with a Dropout layer in between to enhance the model's generalization capability. The output of this sequential component is passed through a dense layer with a Softmax activation to generate frame-level character probabilities.

For training, the model is wrapped with CTC loss layer. The training model takes as input the predicted logits, target label sequences, and their respective lengths, and computes the CTC alignment loss. Since pooling is only performed along the frequency axis, the temporal dimension is preserved, and the effective input length is identical to the original spectrogram time length. Labels are preprocessed into padded index sequences with corresponding true lengths for CTC compatibility. During inference, greedy CTC decoding is applied to the probability outputs to generate character sequences. These settings were optimized through experimentation to ensure robust performance in noisy ASR tasks.

The hyperparameters for W-MSA, SW-MSA, and FG-MSA used in our FG-Swin Transformer model for noisy speech recognition are outlined in Table 4. The MSA mechanism in the Swin Transformer blocks uses 8 attention heads, with key and value dimensions set to 64 per head, and a window size of 7 × 7. In the second Transformer block, a shifted window mechanism enhances contextual understanding by integrating information across overlapping regions. The third Transformer block incorporates frequency-guided MSA, where inputs are weighted based on their frequency importance to emphasize relevant features. Dense layers within the Transformer blocks are configured to reduce dimensions from 512 to 128, with ReLU activation applied.

5.3. Experimental results

Table 5 presents the word recognition accuracy results of the proposed CNN-FG-Swin Transformer model on Aurora-2 dataset under various noise conditions. The results are categorized into three sets: Set A (subway, babble, car, and exhibition noise), Set B (restaurant, street, airport, and station noise), and Set C (subway and street noise). Each column shows recognition accuracy across different SNR levels, ranging from clean conditions to highly noisy environments (−5 dB).

The CNN-FG-Swin Transformer model maintains high accuracy across all noise types, with an average recognition accuracy of 87.19%. As expected, accuracy decreases as noise levels increase, but the model demonstrates strong robustness, particularly in low SNR conditions. At 0 dB, it achieves 77.22% accuracy on average, while at −5 dB, it still performs reasonably well with an average accuracy of 52.07%, proving its effectiveness in highly noisy environments. In Set A, subway noise achieves the highest accuracy at −5 dB with 63.29%, showing the model's adaptability. In Set B, recognition remains more challenging, with restaurant noise scoring 42.28% at −5 dB but improving significantly to 71.95% at 0 dB. Set C maintains a balanced performance, with subway noise reaching 64.30% at −5 dB. These results confirm the CNN-FG-Swin Transformer model's robustness and adaptability, making it a promising approach for ASR in real-world noisy scenarios. A comparison of the CNN-MSA, CNN-Baseline Swin Transformer, and FG-Swin Transformer for noisy speech recognition is also provided in this table, while the parameters of these models are reported in Table 6.

Table 7 presents the recognition accuracy of the CNN-FG-Swin Transformer under various noisy environments (Set A) on isolated words and on the all Aurora-2 dataset, including both continuous and isolated words, in Subway, Babble, Car, and Exhibition

Table 4
Comparison between isolated words and all dataset (continuous speech plus isolated words).

Component	Isolated words	All dataset (Continuous speech plus Isolated words)
Spectrogram input	40 × 98	40 × 300
Conv2D Blocks	Block1: 64 filters, 3 × 3, ReLU, BatchNorm, MaxPool 2 × 2 Block2: 128 filters, 3 × 3, ReLU, BatchNorm, MaxPool 2 × 2 Block3: 256 filters, 3 × 3, ReLU, BatchNorm, MaxPool 2 × 2 Block4: 512 filters, 3 × 3, ReLU, BatchNorm, MaxPool 2 × 2	Block1: 64 filters, 3 × 3, ReLU, LayerNorm, MaxPool 2 × 1 Block2: 128 filters, 3 × 3, ReLU, LayerNorm, MaxPool 2 × 1 Block3: 256 filters, 3 × 3, ReLU, LayerNorm, MaxPool 2 × 1 Block4: 512 filters, 3 × 3, ReLU, LayerNorm, MaxPool 2 × 1
Attention Blocks	Transformer Block 1: Window-Based MSA (W-MSA) - Normalization layer before attention computation. - Heads: 8. - Key Dimension: 64 per head. - Value Dimension: 64 per head. - Query/Key/Value Projection Layers: Linear transformation with output dimension (512). - Window Size: 7 × 7. - Output: Concatenated attention heads, followed by Dense layers with Add (residual connection). - Dense Layers: Units 512 → 128, ReLU activation.	
	Transformer Block 2: Shifted Window-Based MSA (SW-MSA) - Normalization layer before attention computation. - Heads: 8. - Key Dimension: 64 per head. - Value Dimension: 64 per head. - Query/Key/Value Projection Layers: Linear transformation with output dimension (512). - Shifted Window Size: 7 × 7 (offset by half window size for overlap). - Output: Similar concatenated heads with Dense layers and Add operation (residual). - Dense Layers: Units 512 → 128, ReLU activation.	
	Transformer Block 3: Frequency-Guided MSA (FG-MSA) - Normalization layer before attention computation. - Heads: 8. - Key Dimension: 64 per head. - Value Dimension: 64 per head. - Query/Key/Value Projection Layers: Linear transformation with output dimension (512). - Frequency Guidance: Weighted inputs based on frequency importance. - Output: Frequency-guided attention heads with Dense layers and Add operation (residual). - Dense Layers: Units 512 → 128, ReLU activation. MLP Head: Dense layers: Units 512 → 128, Activation: ReLU.	
Sequence modeling	TimeDistributed dense 256 → dropout 0.5 → dense 128 → dropout 0.5	BiLSTM 256 → dropout → BiLSTM 128
Output layer	TimeDistributed, dense num_classes, softmax	Dense num_classes, softmax
Pooling strategy	MaxPooling 2 × 2 (frequency + time)	MaxPooling only along frequency (2 × 1)
Feature flattening	GlobalAveragePooling2D → dense → reshape → TimeDistributed dense	Collapse frequency → BiLSTM
Loss/Training	Categorical crossentropy	CTC loss
Optimizer/LR	Adam, 0.0001	Adam, 0.0001
Batch size/Epochs	32/50	32/50

Table 5
Recognition accuracy results on the Aurora-2 dataset for isolated words using CNN-MSA, CNN-baseline Swin Transformer, and CNN-FG-Swin Transformer.

		Set A				Set B				Set C		Average
		Subway	Babble	Car	Exhibition	Restaurant	Street	Airport	Station	Subway	Street	
CNN-MSA	Clean	96.11	97.13	98.01	97.39	95.50	97.20	97.60	97.32	98.59	97.06	97.09
	20 dB	96.04	95.70	95.48	95.92	96.91	96.43	95.83	95.49	96.92	96.92	96.26
	15 dB	94.03	95.20	94.84	95.21	95.31	94.56	94.30	97.45	95.13	95.13	94.92
	10 dB	93.96	91.47	95.97	92.39	92.01	93.69	94.63	93.80	94.18	93.83	93.59
	5 dB	90.10	89.89	86.01	91.27	88.10	88.10	88.10	88.10	83.23	83.88	87.89
	0 dB	77.58	70.11	77.81	64.93	63.83	70.47	65.16	71.97	77.11	72.76	71.17
	−5 dB	56.64	35.99	45.80	50.63	40.34	50.25	42.61	42.96	52.42	42.08	45.67
	Average	86.21	82.21	84.99	84.39	81.71	84.24	82.46	83.73	85.22	83.38	83.85
CNN-Baseline Swin	Clean	97.79	98.14	98.02	97.68	95.77	97.56	98.06	96.55	98.66	98.28	97.65
	20 dB	97.28	96.56	97.39	97.18	96.53	97.49	97.03	96.76	97.45	97.13	97.08
	15 dB	97.35	97.06	94.77	96.32	95.33	96.85	97.39	96.83	96.12	96.63	96.46
	10 dB	93.76	91.23	93.99	91.55	94.16	92.26	93.47	94.23	90.18	92.16	92.80
	5 dB	91.25	87.10	92.08	91.20	88.32	87.74	84.97	89.72	88.86	90.75	89.00
	0 dB	81.74	70.54	71.59	72.04	69.40	73.91	65.02	73.24	77.18	71.47	72.01
	−5 dB	58.66	40.29	50.04	51.08	39.93	50.97	40.92	43.31	52.95	44.23	47.94
	Average	88.26	82.99	85.41	85.29	82.78	85.25	82.41	84.38	85.91	84.38	84.70
CNN-FG-Swin	Clean	98.12	98.28	98.59	97.96	99.40	98.78	98.09	97.54	98.46	99.07	98.42
	20 dB	98.72	95.99	99.58	96.41	96.78	99.43	99.36	97.46	98.93	98.64	98.13
	15 dB	97.60	97.28	97.81	96.55	96.98	97.13	98.73	98.10	95.03	97.71	97.38
	10 dB	96.04	91.90	96.66	94.86	97.32	95.41	94.70	94.86	90.54	93.76	94.61
	5 dB	95.91	89.96	93.57	97.61	92.15	89.96	85.02	93.17	96.04	91.61	92.50
	0 dB	85.26	71.83	80.49	83.52	71.95	76.49	70.74	73.87	85.64	76.63	77.22
	−5 dB	63.29	46.59	52.51	52.75	42.28	57.17	45.80	46.41	64.30	49.68	52.07
	Average	**90.70**	**84.54**	**87.86**	**88.52**	**85.26**	**87.76**	**84.63**	**85.91**	**89.84**	**86.72**	**87.19**

Table 6
Comparison of CNN-MSA, CNN-Baseline Swin, and FG-Swin Transformer for noisy speech recognition.

Feature/Parameter	CNN-MSA	CNN-Baseline Swin	CNN-FG-Swin
Conv2D block filters	64, 128, 256, 512	64, 128, 256, 512	64, 128, 256, 512
Conv2D block kernel size, Activation	3 × 3, ReLU	3 × 3, ReLU	3 × 3, ReLU
Number of swin transformer Blocks	1	2	3
Attention type	W-MSA	W-MSA+SW-MSA	W-MSA+SW-MSA+FG-MSA
Shift size	0	0, 2	0, 2, 0
Multiple stages (Shifted Windows)	No	Yes	Yes
Window Size	4	4	4
Frequency guidance	No	No	Yes
Hidden units per time step	102	102	102
GlobalAvgPooling2D	Yes	Yes	Yes
TimeDistributed	Yes	Yes	yes
Dropout rate	0.5	0.5	0.5
Optimizer	Adam	Adam	Adam
Learning rate	0.001	0.001	0.001
Batch size	32	32	32
Epochs	50	50	50

noise types, evaluated across multiple SNR levels ranging from clean conditions to −5 dB. The results indicate that the CNN-FG-Swin Transformer consistently delivers higher accuracy in clean and moderately noisy conditions (down to 10 dB), maintaining averages above 94% for isolated words and 97% for all dataset. As the noise level intensifies, particularly at 0 dB and −5 dB, recognition performance decreases, yet remains relatively robust with an overall average of 87.91% in isolated words and 87.01% in all dataset across all SNR levels and noise types. In comparison, other enhancement methods show lower overall performance under the same conditions. For instance, AFE ETSI, CBN, NCNN, and Spark+PBS average ranging from 80.17% to 85.96%, while DWD and Wav2vec.02 trail further behind. These results highlight the CNN-FG-Swin Transformer's superior robustness and adaptability, particularly in real-world noise scenarios, making it highly effective for reliable speech recognition under adverse acoustic conditions.

These results clearly validate the effectiveness and noise resilience of the CNN-FG-Swin Transformer model, showing its suitability for real-world speech recognition applications where strong background noise is often unavoidable.

Table 7
Average recognition accuracy of the CNN-FG-Swin Transformer results (Set A) on the Aurora-2 dataset for isolated words, and all dataset (continuous speech and isolated words).

	Method	SNR	Clean	20 dB	15 dB	10 dB	5 dB	0 dB	−5 dB	Average
isolated words	CNN-FG-Swin Transformer	Subway	98.12	98.72	97.60	96.04	95.91	85.26	63.29	90.70
		Babble	98.28	95.99	97.28	91.90	89.96	71.83	46.59	84.55
		Car	98.59	99.58	97.81	96.66	93.57	80.49	52.51	87.86
		Exhibition	97.96	96.41	96.55	94.86	97.61	83.52	52.75	88.52
		Average	**98.23**	**97.67**	**97.31**	**94.86**	**94.26**	**79.23**	**53.78**	**87.91**
All dataset	CNN-FG-Swin Transformer	Subway	98.83	98.74	98.48	97.58	95.53	85.32	50.17	89.24
		Babble	98.41	97.66	97.41	96.65	91.41	70.36	40.23	84.59
		Car	98.76	98.60	98.43	97.46	94.63	81.55	36.74	86.60
		Exhibition	98.71	98.53	97.97	97.03	93.17	79.23	48.30	87.56
		Average	**98.72**	**98.38**	**98.07**	**97.18**	**93.68**	**79.11**	**43.86**	**87.01**
All dataset	DWD (Soe Naing et al., 2020)		–	–	–	97.54	93.13	75.63	37.29	75.90
	AFE ETSI (Fazel and Chakrabartty, 2011)		99.09	98.15	96.93	93.16	84.36	60.88	28.57	80.17
	Spark and PBS (Fazel and Chakrabartty, 2011)		99.26	98.41	97.18	94.12	84.98	61.38	27.33	80.38
	CBN (Nasersharif and Naderi, 2021)		–	–	–	–	–	–	–	82.83
	NCNN (Rashno et al., 2019)		–	–	–	–	–	–	–	85.96
	Wav2vec.02		–	–	–	–	95.18	75.30	29.97	67.02

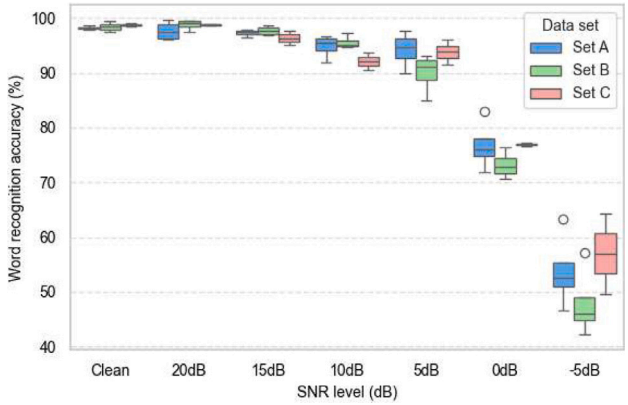


Fig. 4. Box Plot of word recognition accuracy with CNN-FG-Swin Transformer across SNR levels for set A, B and C.

5.4. Statistical analysis

The box plot in Fig. 4 illustrates the variation in word recognition accuracy across different SNR levels in all noise conditions (Subway, Street, etc.) for Sets A, B, and C with the CNN-FG-Swin Transformer, providing insights into the model's performance under varying noise conditions in Aurora-2 dataset. The x-axis represents different SNR levels, while the y-axis indicates the word recognition accuracy. Each box corresponds to a specific SNR level and shows the distribution of accuracy values. The median (middle line in the box) represents the central value of accuracy for each SNR level, providing a clear indicator of the model's typical performance. For high SNR levels, such as Clean and 20 dB, the boxes are tightly clustered with a high median, indicating consistently strong performance with minimal variation. As the noise increases, particularly at low SNR levels such as 0 dB and −5 dB, the median accuracy slightly drops and the IQR widens, yet the model still maintains competitive performance, demonstrating its robustness even under challenging noise conditions. The IQR covers the middle 50% of the data, ranging from Q1 to Q3, indicating a larger spread of accuracy values at lower SNRs. In 5 dB, 0 dB, and −5 dB, the whiskers (lines extending from the box) represent the minimum and maximum values within 1.5×IQR, highlighting the more extreme accuracy values. While some outliers appear outside the whiskers, the overall performance remains strong, demonstrating the model's ability to handle even the most challenging noise conditions effectively.

The Figs. 5–7 illustrate the recognition accuracy across different noise conditions using various model configurations in Aurora-2 dataset. Each graph plots accuracy against SNR values, comparing the baseline W-MSA model, an improved W-MSA+SW-MSA model, and the proposed W-MSA+SW-MSA+FG-MSA model. Across all Figures, the general trend shows an increase in recognition accuracy

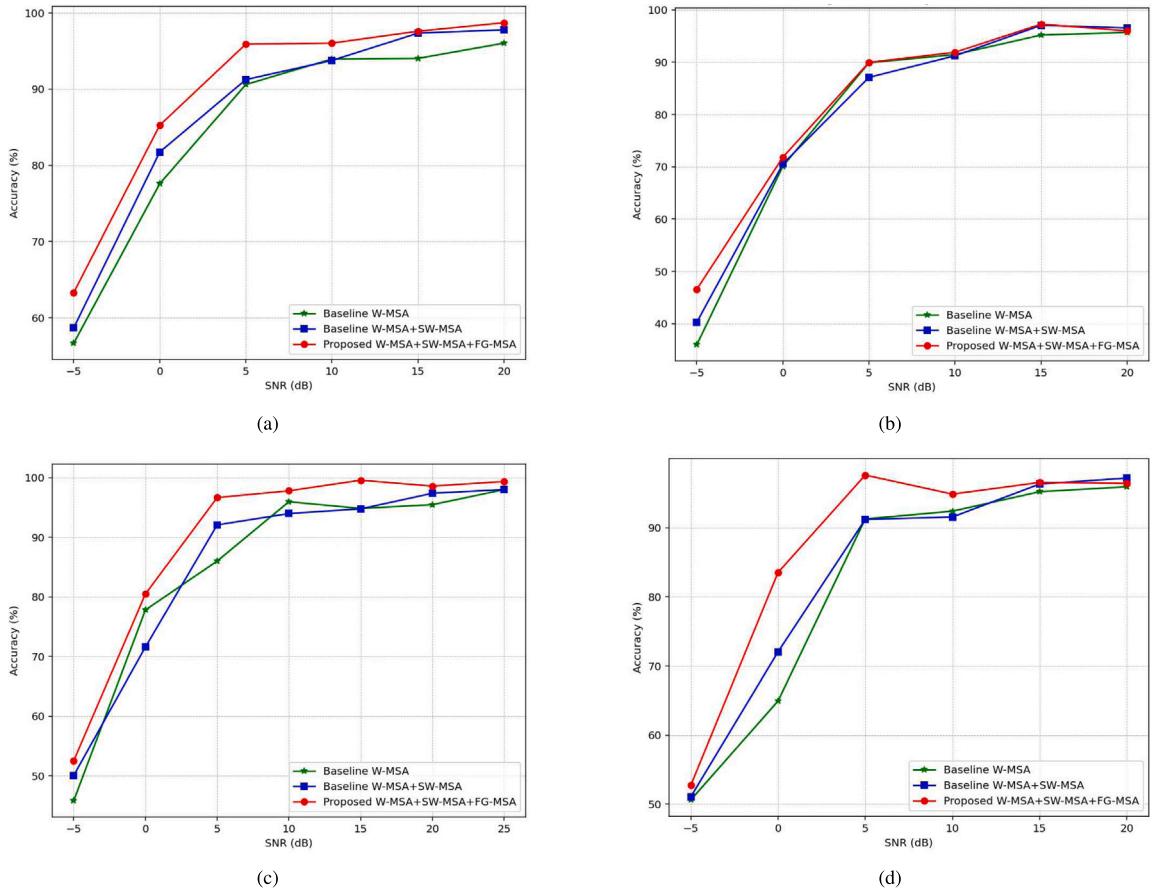


Fig. 5. Recognition accuracy under different additive noise conditions in Set A: (a) Subway noise, (b) Babble noise, (c) Car noise, and (d) Exhibition noise.

as the SNR improves, indicating that higher SNR values contribute to better recognition performance. The proposed model, which integrates FG Swin Transformer, consistently outperforms the baseline models across different noise environments.

For Set A, which includes subway, exhibition, babble, and car noise conditions, the proposed model exhibits superior accuracy, particularly in lower SNR values, suggesting its robustness against highly noisy conditions. Set B, comprising restaurant, station, and airport noise, also follows a similar trend, with the proposed model maintaining a performance advantage over the baseline models. Set C, covering street noise conditions, reaffirms this observation, where the integration of SW-MSA and FG-MSA enhances the model's resilience to convolutive noise distortions. The Figures collectively demonstrate that incorporating FG Swin Transformer further refines speech recognition in noisy environments, especially under challenging SNR conditions. These results validate the efficiency of the proposed enhancements, making them a promising approach for robust ASR in noisy environments.

5.5. Generalizability assessment

To assess the generalizability of the proposed model, we conducted additional experiments on the noisy SC dataset. These evaluations aimed to test the model's robustness under different noise conditions. The results demonstrated strong performance and reinforced the model's capacity to generalize across datasets.

Table 8 presents the accuracy recognition results of the CNN-FG-Swin Transformer model evaluated on the NSC dataset under various noisy conditions. The commands are divided into two categories: main commands (C1) and sub-commands (C2). Across all noise levels, the model demonstrates high recognition accuracy for main commands, with average accuracies of 95.68%, 97.03%, and 96.69% for pink noise, running tap, and white noise, respectively. Overall, the model demonstrates strong robustness to noise. Even at −5 dB, where background noise is very dominant, it maintains impressive accuracy. For main commands, the model achieves 90.59% with white noise and 90.80% with running tap noise at −5 dB, indicating excellent recognition ability despite heavy interference. Even with the more disruptive pink noise, it still reaches 82.00% accuracy. For sub-commands, which are generally more sensitive to noise, the model continues to perform well, dude_miaowing achieves 86.00%, and exercise_bikee maintains 78.84% accuracy at −5 dB. To visualize the model's behavior in diverse acoustic environments, we present a performance comparison under varying noise levels in the NSC dataset, as shown in Fig. 8.

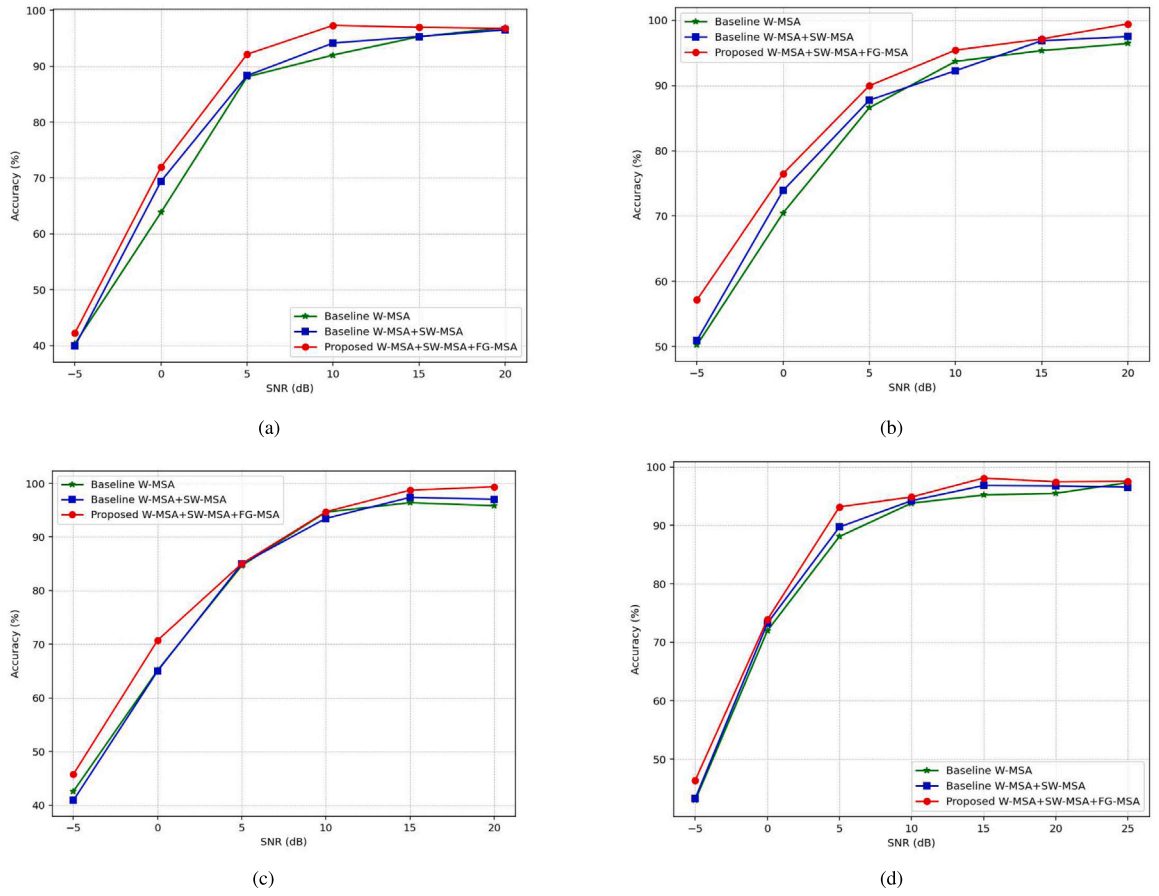


Fig. 6. Recognition accuracy under different additive noise conditions in Set B: (a) Restaurant noise, (b) Street noise, (c) Airport noise, and (d) Station noise.

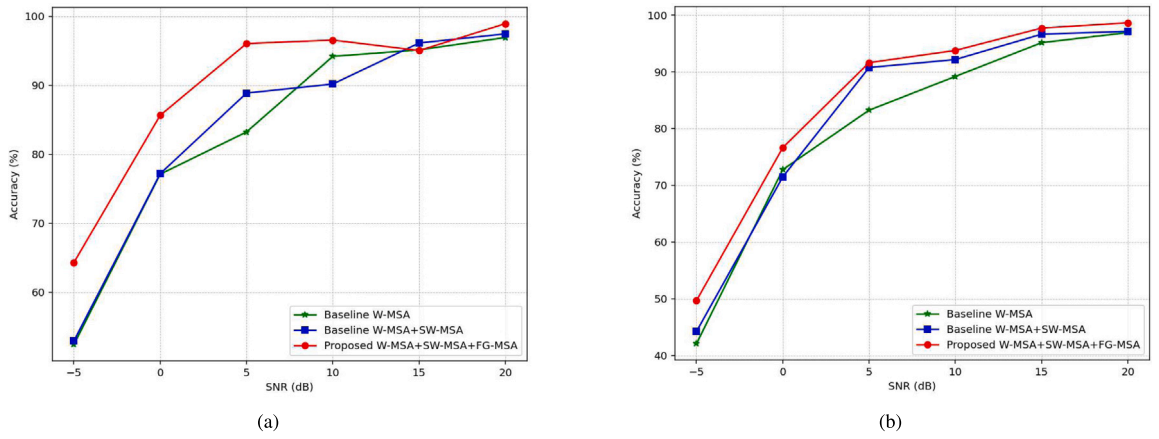
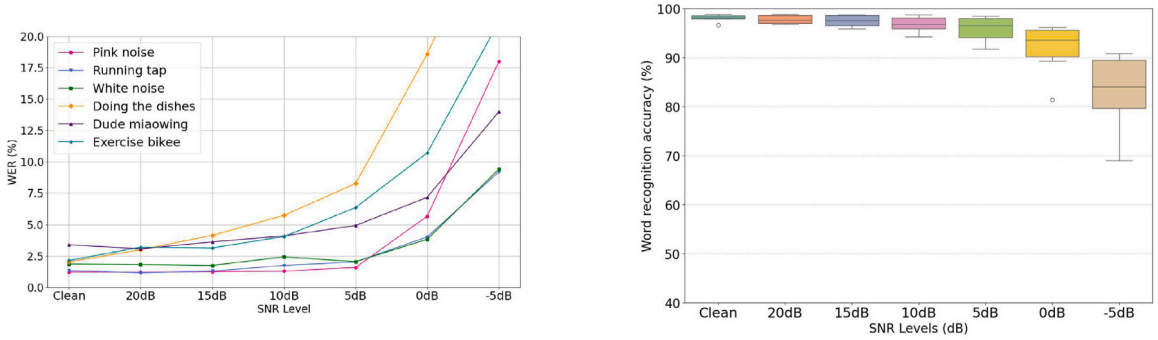


Fig. 7. Recognition accuracy under different additive noise conditions in Set C: (a) Subway noise and (b) Street noise.

Fig. 8(a) illustrates the WER of the CNN-FG-Swin Transformer model under different noise conditions for the main speech commands (C1) and sub-commands (C2) in the NSC dataset. In the main commands, which include pink noise, running tap, and white noise, the results show that the model maintains a very low WER in clean and moderately noisy environments, with error rates staying below 2% for all noise types from clean conditions down to 10 dB. As the noise level increases further, particularly at 5 dB

Table 8
Accuracy recognition results by CNN-FG-Swin Transformer in NSC dataset trained and tested in noisy data.

	Main commands C1			Sub-commands C2		
Noise Type	Pink_noise	Running_tap	White_noise	doing_the_dishes	dude_miaowing	exercise_bikee
Clean	98.78	98.66	98.15	97.98	96.61	97.86
20 dB	98.80	98.85	98.19	97.00	96.93	96.80
15 dB	98.76	98.71	98.26	95.86	96.38	96.86
10 dB	98.71	98.26	97.57	94.26	95.90	95.95
5 dB	98.42	97.98	97.96	91.73	95.07	93.64
0 dB	94.33	95.98	96.17	81.41	92.82	89.29
−5 dB	82.00	90.80	90.59	68.96	86.00	78.84
Average	95.68	97.03	96.69	89.60	94.24	92.74



(a) (b)

Fig. 8. Performance comparison under different noise conditions in the NSC dataset: (a) WER of main commands and Sub- commands; (b) Word recognition accuracy box plot with CNN-FG-Swin Transformer.

and 0 dB, a slight increase in WER is observed, with pink noise causing slightly more errors than running tap and white noise. Even in extremely noisy conditions at −5 dB, the model demonstrates strong robustness: WER remains under 10% for running tap and white noise, and although pink noise leads to a higher WER of around 18%, the model still achieves reasonable recognition accuracy. These results confirm the model's effectiveness in noisy environments, highlighting its suitability for real-world applications where speech recognition must function dependably under various types of background noise.

In the sub-commands, which include Doing the Dishes, Dude Miaowing, and Exercise Bikee, the model also performs well under clean and mildly noisy conditions (from clean to 15 dB), with WER remaining below 5% for all three commands. As the noise level increases (10 dB to 0 dB), the WER gradually rises, especially for Doing the Dishes, which appears more sensitive to noise and reaches about 18% at 0 dB. At the most challenging level of −5 dB, the difference becomes more pronounced: Doing the Dishes sees a sharp increase to over 30% WER, indicating significant difficulty under intense noise. Exercise Bikee also experiences a noticeable rise, reaching about 20%, while Dude Miaowing remains more robust, with WER staying below 15% even at −5 dB. These results suggest that while the model is still capable in noisy conditions, certain sub-commands are more vulnerable to distortion, especially those with more complex or ambiguous acoustic patterns. Nevertheless, the model retains reasonable recognition ability even under severe noise, affirming its general resilience.

The box plot in Fig. 8(b) illustrates the word recognition accuracy of the CNN-FG-Swin Transformer across various SNR levels in the NSC dataset. Each box represents the distribution of accuracy scores, including the median (Q2), interquartile range (IQR) between the first quartile (Q1) and third quartile (Q3), and any outliers. In high SNR conditions (Clean to 15 dB), the boxes are tightly grouped near the upper end (above 95%), indicating high and consistent performance with minimal variation. As noise increases (10 dB to 0 dB), the boxes begin to widen slightly, showing increased variability in accuracy. At −5 dB, the spread becomes much larger, with the lower quartile dropping below 80% and some values nearing 70%, reflecting the greater impact of noise on recognition accuracy. Despite this degradation, the model continues to achieve relatively high performance even in severe noise, confirming its robustness and effectiveness for real-world speech recognition in challenging acoustic environments.

Regarding the higher performance observed on the NSC dataset compared to Aurora-2 especially in low SNR conditions such as −5 dB, one contributing factor is the sampling rate. The 16 kHz resolution used in NSC allows for better capture of phonetic transitions and fine-grained acoustic details than the 8 kHz used in Aurora-2, which facilitates more accurate recognition of spoken commands. In contrast, Aurora-2 presents a significantly more challenging scenario, as it includes realistic urban background noises (e.g., street, subway, station, airport) that are much more acoustically complex than synthetic noises such as pink or white noise.

Table 9
Comparison of different models highlighting the importance of adding CNN in the proposed model on subway noise (−5 dB) Test A.

	Model	Acc. (%)	WER (%)
Isolated dataset	Baseline Swin	54.66	45.34
	FG-Swin + CNN (2 Blocks)	44.36	55.64
	FG-Swin + CNN (3 Blocks)	59.66	40.34
	FG-Swin + CNN (4 Blocks)	**63.29**	**36.71**
	FG-Swin +CNN (5 Blocks)	60.54	39.46
	Baseline Swin (W-MSA+SW-MSA) + CNN (4 Blocks)	58.66	41.34
	W-MSA + CNN (4 Blocks)	56.64	43.36
	(W-MSA+SW-MSA+FG-MSA) +CNN (4 Blocks)	**63.29**	**36.71**
All dataset	CNN-FG-Swin+CTC	47.18	52.82
	CNN-FG-Swin+CTC+BiLSTM	50.17	49.83

These real-world noise conditions, particularly under lower SNR levels, make it substantially more difficult to separate speech from background interference compared to the NSC dataset.

5.5.1. Ablation study

Table 9 compares different models to highlight the importance of adding CNN in the proposed FG-Swin Transformer model under Subway Noise (−5 dB) Test A conditions in Aurora-2 dataset. The results show that increasing the number of CNN blocks improves accuracy and reduces the WER. For instance, in the FG-Swin Transformer + CNN model, increasing CNN blocks from 2 to 3 significantly boosts accuracy from 44.36% to 59.66% and reduces WER from 55.64% to 40.34%. Further increasing to 4 blocks results in the highest accuracy 63.29% and the lowest WER 36.71%, confirming that adding more CNN blocks enhances robustness against noise. To determine whether performance continues improving beyond 4 blocks, we extended the ablation study by testing a configuration with 5 CNN blocks. The results showed a slight decrease in accuracy 60.54% and no further improvement in WER 39.46%, suggesting that the benefit saturates or may even degrade beyond 4 blocks. This justifies selecting 4 CNN blocks as the optimal configuration for balancing performance and model complexity.

However, the Baseline Swin Transformer model, without any CNN augmentation or frequency-guided attention, achieves an accuracy of 54.66% and a WER of 45.34%. This relatively modest performance demonstrates the limitations of the vanilla Swin Transformer in extreme noise environments, where input features are highly degraded. To explore the contribution of convolutional feature extraction, four CNN blocks are added before the Transformer layers in the Baseline Swin + CNN (4 Blocks) configuration. This results in a modest improvement as accuracy increases to 58.66%, and WER decreases to 41.34%. This suggests that CNNs help capture local and hierarchical representations, which improve noise robustness.

As shown in Table 9, the W-MSA + CNN (4 Blocks) variant further investigates the role of self-attention by incorporating only W-MSA, without the inclusion of shifted windows or frequency-guided attention. This model achieves an accuracy of 56.64% and a WER of 43.36%, indicating that while W-MSA contributes positively, the absence of shifted attention reduces its effectiveness in low-SNR conditions.

The best performance is achieved by the FG-Swin Transformer + CNN (4 Blocks) configuration, which incorporates W-MSA, SW-MSA, and FG-MSA, along with the same four CNN blocks. This model achieves 63.29% accuracy and the lowest WER of 36.71%, demonstrating that combining frequency-aware attention with local CNN features leads to substantial gains in robustness. The inclusion of FG-MSA allows the model to better capture frequency-domain dependencies that are essential in speech recognition tasks affected by noise.

This ablation study confirms that each architectural component contributes incrementally to model performance. The CNN layers improve low-level feature extraction, while FG-MSA significantly strengthens the model's capacity to handle frequency distortions introduced by noise. The FG-Swin Transformer with CNN emerges as the most effective design, offering superior accuracy and resilience in highly noisy environments. A BiLSTM layer is crucial in speech recognition because it captures temporal context in both directions past and future which is essential for understanding sequences of sounds. Speech is inherently sequential, and the correct interpretation of a sound or word often depends on what comes before and after it. Although the Transformer's self-attention mechanism captures *global* bidirectional dependencies, a BiLSTM layer is incorporated to reinforce the modeling of *local* temporal context, which is essential for accurately decoding continuous speech signals. In models trained with CTC loss, BiLSTM layers provide richer sequence representations, enabling better alignment between input audio frames and target text. Removing BiLSTM can make the model faster, but it significantly reduces accuracy as shown in Table 9, as the model loses the ability to fully understand long-term dependencies in the audio signal. Thus, BiLSTM plays a key role in ensuring robust and accurate transcription in speech recognition systems.

Table 10
Performance accuracy (%) of the proposed CNN-FG-Swin Transformer model compared to state-of-the-art methods under different noise types (Test A) in the Aurora-2 dataset for isolated words and all dataset, while the referenced compared works are evaluated on the full dataset including continuous speech.

	Work	Year	Model	Clean	20 dB	15 dB	10 dB	5 dB	0 dB	−5 dB
Subway noise	Soe et al. Soe Naing et al. (2020)	2020	DWD	–	–	–	98.04	95.36	80.20	44.55
	Naser Sharif et al. Nasersharif and Naderi (2021)	2021	CBN	–	95.30	94.22	91.34	89.39	56.86	26.89
	Fazel et al. Fazel and Chakrabartty (2011)	2012	AFE ETSI	99.08	97.91	96.41	92.23	83.82	61.93	30.86
	Fazel et al. Fazel and Chakrabartty (2011)	2012	SPARK AND PBS	99.36	98.10	96.41	92.94	82.87	59.26	27.97
		2025	End-to-end Wav2vec2.0 (All)	–	–	–	–	96.70	80.62	39.56
		2025	CNN-Baseline Swin (Isolated)	97.79	97.28	97.35	93.76	91.25	81.74	58.66
	Proposed	**2025**	CNN-FG-Swin isolated words	**98.12**	**98.72**	**97.60**	**96.04**	**95.91**	**85.26**	**63.29**
			CNN-FG-Swin for all dataset	**98.83**	**98.74**	**98.48**	**97.58**	**95.53**	**85.32**	**50.17**
Babble noise	Soe et al. Soe Naing et al. (2020)	2020	DWD	–	–	–	97.64	91.38	69.70	33.86
	Fazel et al. Fazel and Chakrabartty (2011)	2012	AFE ETSI	99.00	98.31	96.89	92.35	81.08	51.90	19.71
	Fazel et al. Fazel and Chakrabartty (2011)	2012	SPARK AND PBS	99.12	98.70	97.64	95.37	86.61	58.12	21.58
		2025	End-to-end Wav2vec2.0 (All)	–	–	–	–	94.31	71.13	23.88
		2025	CNN-Baseline Swin (Isolated)	98.14	96.56	97.06	91.23	87.10	70.54	40.29
	Proposed	**2025**	CNN-FG-Swin isolated words	**98.28**	**95.99**	**97.28**	**91.90**	**89.96**	**71.83**	**46.59**
			CNN-FG-Swin for all dataset	**98.41**	**97.66**	**97.41**	**96.65**	**91.41**	**70.36**	**40.23**
Car noise	Soe et al. Soe Naing et al. (2020)	2020	DWD	–	–	–	97.79	93.23	76.40	25.05
	Fazel et al. Fazel and Chakrabartty (2011)	2012	AFE ETSI	99.05	98.48	97.58	95.29	88.49	66.42	30.84
	Fazel et al. Fazel and Chakrabartty (2011)	2012	SPARK AND PBS	99.19	98.69	98.03	95.47	86.76	71.28	34.54
		2025	End-to-end Wav2vec2.0 (All)	–	–	–	–	94.91	73.23	26.07
		2025	CNN-Baseline Swin (Isolated)	98.02	97.39	94.77	93.99	92.08	71.59	50.04
	Proposed	**2025**	CNN-FG-Swin isolated words	**98.59**	**99.58**	**97.81**	**96.66**	**93.57**	**80.49**	**52.51**
		2025	CNN-FG-Swin for all dataset	**98.76**	**98.60**	**98.43**	**97.46**	**94.63**	**81.55**	**36.74**
Exhibition noise	Soe et al. Soe Naing et al. (2020)	2020	DWD	–	–	–	96.70	92.56	76.20	45.70
	Fazel et al. Fazel and Chakrabartty (2011)	2012	AFE ETSI	99.23	97.90	96.82	92.78	84.05	63.28	32.86
	Fazel et al. Fazel and Chakrabartty (2011)	2012	SPARK AND PBS	99.38	98.15	96.64	92.69	81.67	56.77	25.24
		2025	End-to-end Wav2vec2.0 (All)	–	–	–	–	94.81	76.22	30.37
		2025	CNN-Baseline Swin (Isolated)	98.02	97.39	94.77	93.99	92.08	71.59	50.04
	Proposed	**2025**	CNN-FG-Swin isolated words	**97.96**	**96.41**	**96.55**	**94.86**	**97.61**	**83.52**	**52.75**
		2025	CNN-FG-Swin for all dataset	**98.71**	**98.53**	**97.97**	**97.03**	**93.17**	**79.23**	**48.30**

5.6. Comparison with existing work

Table 10 present a comparative analysis of the proposed CNN-FG-Swin Transformer model against state-of-the-art methods for noisy speech recognition on isolated words and on the all Aurora-2 dataset, including both continuous and isolated words under various noise conditions. In Subway Noise, the CNN-FG-Swin Transformer outperforms all existing models across different noise levels. Compared to traditional models such as DWD, CBN, SPARK, PBS, and even recent models like Wav2Vec 2.0 with fine-tuning in Aurora2 dataset, the proposed model achieves superior accuracy, particularly under extreme noise conditions such as −5 dB, highlighting its robustness and effectiveness in highly noisy environments. In Babble Noise, where a similar trend is observed, the CNN-FG-Swin Transformer consistently outperforms other models across all noise levels. While the CNN-Baseline Swin Transformer demonstrates competitive accuracy, the integration of frequency-guided self-attention in the proposed model significantly enhances recognition, especially under low SNR conditions. In Car Noise, the CNN-FG-Swin Transformer again surpasses baseline methods, achieving superior accuracy, especially at challenging SNR levels from 5 dB to −5 dB, where it retains a clear advantage over conventional techniques such as AFE ETSI, SPARK, PBS, and even the more recent Wav2Vec 2.0. This further underscores the robustness of the proposed model in diverse acoustic environments. In Exhibition Noise, the CNN-FG-Swin Transformer maintains its dominance by delivering higher accuracy across all SNR levels. Its advanced self-attention mechanisms and frequency-guided adaptation enable more effective noise suppression compared to both traditional and recent state-of-the-art models. The comparative analysis across the four noise environments demonstrates that the CNN-FG-Swin Transformer is the most robust and accurate model for noisy speech recognition on the Aurora-2 dataset. Its consistent performance improvements over baseline methods and modern

Table 11
WER of deep models trained on clean data and tested on noisy data from the SC dataset at -9 dB across all noise types.

Model	Year	WER (%)
End-to-end Caps-inputvec4 (Bae and Kim, 2018)	2018	47.3
End-to-end Caps-channel32 (Bae and Kim, 2018)	2018	47.4
End-to-end Caps-outputvec4 (Bae and Kim, 2018)	2018	44.7
LSTM (Pervaiz et al., 2020)	2020	69.09
DNN (L = 1) (Pervaiz et al., 2020)	2020	38.63
DNN (L = 2) (Pervaiz et al., 2020)	2020	49.13
CNN-Baseline Swin	2025	40.82
CNN-FG-Swin	**2025**	**36.20**

architectures such as Wav2Vec 2.0 with fine-tuning, confirm the effectiveness of its novel design in handling diverse real-world noise conditions.

Table 11 presents a comparison of WER performance for various DL models trained on clean data and tested on extremely noisy conditions (−9 dB SNR) using the SC dataset. The Table includes models developed between 2018 and 2025, spanning different architectures, such as end-to-end models with capsule networks (Caps-inputvec4, Caps-channel32, and Caps-outputvec4), LSTM, DNN, and Transformer-based models. Traditional models like LSTM and capsule networks show relatively high WERs, with LSTM reaching the highest error rate of 69.09%, indicating poor performance under severe noise. The CNN-FG-Swin Transformer show significant improvements achieves the lowest WER of 36.20%, outperforming both the baseline Swin Transformer (40.82%) and all other earlier models. This demonstrates the superior robustness and generalization capability of the CNN-FG-Swin Transformer when recognizing speech under extreme noise conditions, making it a promising solution for real-world noisy environments. Such performance makes the CNN-FG-Swin Transformer a strong candidate for real-world speech recognition applications where background noise is a significant challenge.

6. Conclusion

This research introduced the FG-Swin Transformer for noisy speech recognition, leveraging the unique ability of frequency-guided mechanisms to enhance feature extraction in the spectral domain. The FG-Swin Transformer architecture integrated frequency-guided attention with Swin Transformer's window-based self-attention mechanisms to address challenges associated with noise distortion in speech signals. Experimental results on the Aurora-2 dataset and NSC dataset demonstrated that the FG-Swin Transformer achieved superior performance compared to baseline models across various noise conditions, including extreme environments such as −5 dB. The integration of frequency-guided attention enabled the model to isolate relevant frequency bands more effectively while suppressing noise, resulting in robust and accurate recognition. The model's performance underscores the importance of combining domain-specific insights, such as frequency guidance, with advanced attention mechanisms to improve speech recognition in noisy environments. By comparison, baseline models lacked the ability to dynamically adapt to the frequency-dependent characteristics of speech signals, highlighting the significance of the proposed approach.

Future research could explore the application of the FG-Swin Transformer to other datasets such as CHiME-4, other speech recognition tasks, and further optimizations for real-time implementation. The advancements introduced by this architecture clear the way for additional robust and efficient noisy speech recognition systems, addressing key challenges in real-world scenarios.

CRediT authorship contribution statement

Noussaiba Djeffal: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Djamel Addou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hamza Kheddar:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sid Ahmed Selouani:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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